

comes with a stereo line in, which is good for recording high-quality stereo audio to the device but it stores this audio in an uncompressed WAV format, thereby taking up enormous amounts of memory. The Nomad Jukebox and the Nomad II MG were the only players that were capable of firmware upgrades, as well as the only two capable of playing WMA files. The Philips Rush and the Casio WMP-1V weren't even capable of playing any VBR (Variable Bit Rate) or more than 128K MP3 files, although all others including the Philips eXpanium played VBRs and other MP3 bitrates just fine. The eXpanium plays multichannel CD-Rs and RWs just fine.

The Nomad Jukebox is a real battery guzzler; it uses four custom made AA-sized rechargeable batteries that can be charged right inside the player itself. While it also comes with an extra set of rechargeable batteries, thereby extending the battery life to a certain extent, the pain of recharging both pairs separately still remains. The best in terms of batteries is the Rio 300, which operates off just one AA-sized alkaline battery and works just fine for more than six hours. Even the Nomad comes with two AAA-sized rechargeable batteries that can be charged using the charging cradle that also doubles up as the data cradle, eliminating hassles with cables. All other portable players use standard batteries.

In terms of memory upgradeability, all the 32 MB players support upgrades of up to 64 MB, with the Nomad even capable of going up to 96 MB with a 64 MB Smart-Media card. The Philips Rush can be upgraded by putting in another memory card of 64 MB, but this is rather pointless as you would first have to remove the existing card, since the base unit doesn't come with any built-in memory, just a 32 MB Smart Media card.

Audio tests

When you're on the move, your headphones need to be loud to be audible, just the regular audio level won't do. All the MP3 players ship with poor quality headphones, as a result of which you miss out on so much fidelity while listening to high quality audio. So the first step is generally to grab a good pair of headphones. Good headphones require more power to drive them, so your MP3 player needs to be more powerful. The louder an MP3 player is, the larger and more power hungry headphones it can drive.

Real quality means the MP3 player should be capable of producing audio just

Variable Bit Rates

Variable Bit Rates (VBR) is a method of encoding that makes allowances for the variation in slices of sound that an MP3 file is made up of. It ensures consistently high audio quality throughout an encoded file by making intelligent bit-allocation decisions while encoding. The encoder allocates a higher bitrate in complex sections to ensure audio quality. In simpler sections such as those containing silence or a relatively narrow range of frequencies, it allocates fewer bits. Thus, VBR is

ideal for when consistent audio quality is more important than the encoded MP3 file size.

Several MP3 players may have trouble reporting the duration of VBR-encoded files and/or not have proper support for it. Most new releases of MP3 playback software can play back MP3 encoded files. The popular Winamp player has had support for it since the release of version 2.05. However, some portable MP3 players may not have support for VBR.

the way it is, and as accurately as possible. The audio playback should be consistent and without any jerks or skips caused due to movements or other such common activities that an MP3 player is subjected to. The device should be capable of offering a sufficient amount of playtime. The playtime is the maximum amount of time for which it can play near-CD quality audio using the lowest bit rate in any audio format it supports. For MP3s, near-CD

both high frequency audio and low frequency audio and even though you can't call them loud, they were enough to drive the headphones to a perfectly acceptable level, without any distortion whatsoever.

When it comes to audio reproduction accuracy, almost all the MP3 players except the Philips Rush performed just fine. The Philips Rush doesn't support 256 Kbps MP3s, and hence the accompanying interface. The Real Jukebox re-encodes all such MP3s to 96 Kbps CBR MP3s and transfers these to the player, hence losing out on a lot of quality. Both the D-Link and the eXpanium were slightly heavier on the bass reproduction all around, even with the bass expander and the equalizer turned off. The Casio MP3 watch failed to run any of these tests entirely as it didn't even support any conversion on the interface for re-encoding the MP3s into a format compatible with the device.

All the MP3 players, including the Jukebox and the eXpanium were flawless when it comes to audio consistency. None of them skipped a beat. We tried everything imaginable to make these devices skip; from rattle moves to the hip swings and even bouncing them off the floor, but they just refused to skip a beat.

The **Nomad Jukebox** was obviously the best when it came to playtime. With its enormous hard disk storage, it can pack in around 200 hours of near-CD quality music in the WMA format. The second was the eXpanium, which can play a CD full of MP3s and the third was the Nomad II MG with WMA 64 Kbps audio. This player is capable of carrying 136 minutes of near-CD quality audio, which is the best amongst all the solid-state players we tested. All the other 32 MB players are capable of doing 34 minutes of playtime at 128 Kbps MP3.



Philips eXpanium: Its one line LCD display shows only the song number

quality is 128 Kbps and for WMA the very same quality can be achieved at half the bit rate, hence doubling the total playtime.

When it comes to audio power, only two MP3 players were really capable of getting our big reference headphones to max out. The D-Link DMP 100 and the Philips eXpanium both offered loud and clear sound, although the D-Link's extra bass equalizer settings were a little distorted at high volumes. The Rio 300, Nomad and the Nomad II MG were very crisp in